# INTERIM FEDERAL SPECIFICATION

## MAT, FLOOR, AND MATTING, PILE SURFACE, ENTRANCE WAY TYPE

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

### 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for non-skid and non-slip floor mats and matting with pile yarn surface for use in entrance ways, corridors, lobbies, hallways, elevators and similar areas, to reduce tracking of dirt, water, snow and grit into the interior of buildings, and to help prevent slips and falls, to protect floors and to reduce noise.

1.2 Classification.

1.2.1 Types and classes. The mats and matting shall be of the following types and classes as specified (6.2).

Type I - Mats or matting with pile surface of tufted carpet construction.
Type II - Mats or matting with pile surface and solid backing.

Class 1 - Solid vinyl backing
Class 2 - Solid rubber backing

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

Federal Specifications:

CCC-T-191 - Textile Test Methods.
MMM-A-260 - Adhesive, Water-Resistant (For Sealing Waterproofed Paper).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 501 - Floor Coverings, Resilient, Nontextile: Sampling and Testing.
Fed. Test Method Std. No. 601 - Rubber, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.



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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C.)

American Society for Testing and Materials
(ASTM) Standards:

D-418-42 - Woven Pile Floor Coverings.
D-925-63 - Contact and Migration Stain of Vulcanized Rubber in Contact with Organic Finishes.
D-1486-57T - Tufted Pile Floor Coverings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa.).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 Preproduction sample. When specified (6.2), a preproduction sample of the mat or matting, measuring one foot square, shall be submitted to the contracting officer for approval.

3.2 Standard sample. When a standard sample is available, the finished mat or matting shall match the standard sample for color and appearance and shall be equal to or better than the standard sample, with respect to all characteristics for which the standard sample is referenced (6.5).

3.3 Bid sample. When required (6.2), bid samples shall be submitted to the contracting officer as specified in the Invitation for Bid.

3.4 Material.

3.4.1 Pile yarn. The pile yarn shall be at least 2-ply and made of 100 percent nylon or wool or a blend of these fibers as specified in the contract or order (6.2).

3.4.1.1 Nylon. Nylon shall be carpet type fiber, with average fiber size of 15 denier or coarser. The finished yarn shall contain not more than 2.0 percent chloroform soluble material when tested as specified in 4.3.1.

3.4.1.2 Wool. Wool shall be thoroughly scoured carpet type fiber which has never been reclaimed from any woven, tufted, knitted or felted products. The wool yarn shall contain a minimum of 95 percent wool based on the dry weight of the specimen when tested as specified in 4.3.1.

3.4.2 Backing material.

3.4.2.1 Type I. The basic or first backing material shall be a woven 100 percent polypropylene fabric not less than 3.2 ozs. sq. yd. or a nonwoven fabric, consisting of not less than 75 percent polypropylene fibers with a balance of cellulosic fibers, not less than 4.0 ozs. sq. yd. The reinforcement or second backing shall be woven fabric, made of jute or plasticized kraftcord yarns, not less than 5.0 ozs. sq. yd.



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3.4.2.2 Type II, classes 1 and 2. The base cloth for type II, classes 1 and 2, mats or matting shall be a cotton or jute woven cloth, 6.0 ozs. sq. yd. minimum, with the yarns per inch a minimum of 12 in the warp and 10 in the filling. The solid backing for type II, class 1, mats or matting shall be vinyl as specified in 3.4.2.2.1. The solid backing for type II, class 2, mats or matting shall be rubber as specified in 3.4.2.2.2.

3.4.2.2.1 Vinyl backing (class 1). The solid backing for type II, class 1, mats or matting shall be suitably compounded virgin polymer or copolymer of vinyl chloride resin, plasticized with phosphate or phthalate ester plasticizers only. The compound shall be uniform, free from fiber and objectionable odor. Virgin polymer shall be defined as a polymer or copolymer that has not been processed into a finished product prior to use in the specified matting. The backing shall weigh not less than 6.0 lbs. sq. yd. The thickness shall be 0.110 (+0.030,-0.015) inch. When subjected to the accelerated weathering test specified in Table III, the backing shall not crack, become stiff and brittle or soft and tacky and shall not change color appreciably when compared to the unexposed sample after 100 hours in the weatherometer.

3.4.2.2.2 Rubber backing (class 2). The solid backing for type II, class 2, mats or matting shall be made from synthetic rubber or a mixture of synthetic and reclaimed rubber. The compound shall be free from objectionable odor. The backing shall weigh not less than 4.75 lbs. sq. yd. The thickness shall be 0.130 (+0.050,-0.040) inch. When subjected to the accelerated aging test specified in Table III, the backing shall not become brittle, nor tacky and shall not show any evidence of deterioration.

3.4.3 Back coating (type I). Back coating for type I mats or matting shall consist of a latex compound applied to the basic backing fabric (3.4.2.1) and a pure, natural (not synthetic) latex applied in laminating the second backing fabric (3.4.2.1).

3.5 Construction and physical requirements.

3.5.1 Type I. The mats or matting shall be of tufted carpet construction and shall have a plain overall pattern of loop pile face. The pile yarn shall be as specified in the contract or order and shall have not less than 3.0 turns per inch. The tufts shall be needled into the basic backing fabric (3.4.2.1). A backing reinforcement (3.4.8.1) shall be applied to the back or floor side of the mats or matting. Not less than 15 ozs. sq. yd. of the back coating (3.4.3) shall be present on the basic backing and not less than 8 ozs. sq. yd. of back coating (3.4.3) shall be present in laminating the second backing so applied that the second backing is completely coated with the natural latex compound to render it skid resistant. Binding of edges shall be as specified in 3.5.1.1. The finished type I mats or matting shall conform to the physical requirements specified in table I.

TABLE I - Physical requirements of type I mats or matting

| Characteristic | Requirement |
|---|---|
| Tufts per square inch, minimum | 38.0 |
| Weight, pile yarn, ozs. sq. yd., minimum | 20.0 |
| Weight, overall, ozs. sq. yd., minimum | 50.0 |
| Pile height, inches: | |
| minimum | 0.250 |
| maximum | 0.350 |
| Resistance to wet-method cleaning | Pass 1/ |
| Flame resistance, longest diameter of charred area, inches, maximum | 2.0 |

1/ There shall be no excessive tearing or removal of the pile surface when subjected to the wet-method cleaning test.

3.5.1.1 Binding of edges (type I). Unless otherwise specified (6.2), all edges shall be bound with 3/4 inch woven cotton tape having not less than 100 total ends and 27 picks per square inch and weighing not less than 0.135 oz. per linear yard. The color of the tape shall be compatible with the color of the pile. Unless otherwise specified (6.2), the tape shall be sewn to the edges with nylon thread at 3 3/4 stitches per inch, and the top thread shall be nylon 52 clear monofilament.



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3.5.2 Type II, classes 1 and 2. Type II, classes 1 and 2, shall consist of a pile yarn surface using pile fiber as specified in the contract or order (see 6.2). The pile yarn shall be woven or needled into base cloth (3.4.2.2) and fused to vinyl backing (3.4.2.2.1) for type II, class 1, or cured or vulcanized to rubber backing (3.4.2.2.2) for type II, class 2, mats or matting. Unless otherwise specified (6.2), the pile surface shall be made of cut or loop pile yarn at the option of the contractor. The pile yarn with base cloth shall cover the entire surface of the solid backing with exceptions as specified in 3.6.2. The fusing, curing or vulcanizing shall result in firm and secure attachment of the pile yarn and base cloth to the solid backing. The mats or matting shall be non-skid and the edges shall be evenly cut. The fusing, curing or vulcanizing operation shall not have any material effect on the properties of the solid backings as specified in 3.4.2.2.1 and 3.4.2.2.2. The backing shall be resistant to curl when tested as specified in 4.4. The finished type II, classes 1 and 2, mats or matting shall conform to the physical requirements specified in table II.

TABLE II - Physical requirements of type II mats or matting

| Characteristic | Requirement | |
|---|---|---|
| | Class 1 | Class 2 |
| Tufts per square inch, minimum | 42.0 | 33.0 |
| Weight, pile yarn, ozs. sq. yd., minimum 1/ | 23.0 | 23.0 |
| Weight, overall, lbs. sq. yd., minimum | 7.75 | 6.25 |
| Thickness, overall, inches, minimum 2/ | 0.43 | 0.50 |
| Pile height, inches, minimum 3/ | 0.37 | 0.40 |
| Flexibility: | | |
| at 70° F. | 4/ | 4/ |
| at 30° F. | 4/ | 4/ |
| Resistance to wet-method cleaning | Pass 5/ | Pass 5/ |
| Flame resistance, longest diameter of charred area, inches, maximum | 2.0 | 2.0 |

1/ May be accepted based on contractor's certificate of compliance.
2/ Brush up pile to take measurement.
3/ Measure from top of base cloth along edge. Brush up pile to take measurement.
4/ The specimen shall not crack, flake, craze or break or show any other indication of failure.
5/ There shall be no excessive tearing or removal of the pile surface when subjected to the wet-method cleaning test.

3.6 Nosing and borders.

3.6.1 Type I. When nosing is specified (6.2), the nosing on the edges so designated shall be molded or extruded rubber. The nosing shall be 2 3/4 to 3 inches wide, of which the lower lip that goes under the matting shall be 1 3/8 to 1 1/2 inches long and 1/8 inch thick and the remainder that extends beyond the matting shall taper from 1/16 inch at the front edge to 1/4 inch, so as to be approximately level with the top of the pile surface. The latter portion shall have a beveled recess for insertion of the bound edges specified. The underside of the nosing shall be ribbed to prevent slippage. The nosing shall be securely sewn to the inserted carpet edge with sufficient rows of double stitching, using a 4 cord No. 16 cotton thread or equivalent. Required nosings shall lay firmly and flat on the floor surface after attachment to the mats.

3.6.2 Type II, classes 1 and 2. The matting when furnished in rolls shall have a border (without pile and base cloth on surface) on each side measuring in width 1(+1/4,-1/16) for class 1 and 1(+5/16, -1/8) inch for class 2. When special cut sizes are ordered, edges may be required to be flush without nosing or border or with nosing or borders as specified in the contract or order (6.2). When nosing or borders are specified for special cut sizes, they shall measure in width 1(+5/16,-1/16) inch for class 1 and 1(+5/16,-1/8) inch for class 2. Thickness of nosing or borders shall not be less than, but may exceed the thickness of the solid backing materials, with a maximum thickness to be approximately level with the top of the pile surface. Nosings and borders on the finished mats or matting shall lay firmly and flat on the floor surfaces and shall be resistant to curl when tested as specified in 4.4.

3.7 Size. The matting shall be furnished in rolls or in special cut sizes as mats in accordance with dimensions as specified in the contract or order (6.2). Tolerances for the width of mats and matting and for the length of mats shall be not more than 1 percent less than specified for type I and not more than 3/4 inch for type II. Examination of rolls of matting for length shall be as specified in 4.3.2.3.



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3.8 Color and matching. The color shall be as specified in the contract or order (6.2). The color specified shall apply to the color of the pile surface. For type II mats and matting, the color of the vinyl backing for class 1 shall be black, except that brown will be allowed for a beige pile surface, and the color of the rubber backing for class 2 shall be compatible with the color of the pile surface. The color of the pile surface shall match the standard sample, when available, under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° K. when tested as specified in 4.5.13.

3.8.1 Colorfastness. The pile surface shall show good colorfastness to water and wet-method cleaning and shall show a rating of good for light colors after 20 standard fading hours (2000 Langleys) and for dark colors after 40 standard fading hours (4000 Langleys). For type II, class 1, vinyl and class 2, rubber, backing materials, there shall be no evidence of migration of color nor shall the backing contain any material that may stain, injure, or mar the appearance of floors.

3.8.2 Contact and migration stain (type II). When specified in the contract or order (6.2), type II mats or matting shall meet requirements specified (6.2) for contact and migration stain of the backing in contact with organic finishes or flooring when tested as specified in 4.4.

3.9 Shrinkage (type I). The shrinkage shall not exceed 3.0 percent in either the length or width, when tested as specified in 4.4, for type I mats or matting.

3.10 Moth repellency. Unless otherwise specified (6.2), a moth repellent compound shall be properly applied to the wool pile component of mats or matting and shall have an insect resistance classification of not less than "resistant" when tested as specified in 4.4.

3.11 Fiber identification. Mats and matting shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Workmanship. Mats and matting shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, inspection shall be made of the completed end item for conformance with all provisions of this specification.

4.3 Inspection for acceptance. Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Testing of components and materials. In addition to the quality assurance provisions of the subsidiary specifications and standards, testing shall be performed on the components and materials listed in table III. The sample size shall be as specified in table IV. The lot shall be unacceptable if one or more sample units fail to meet any specified requirement. The lot size shall be expressed in terms shown in table III. The sample unit for each component or material shall be as specified in table III.



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TABLE III - Component testing

| Component and lot size expressed in terms of | Characteristic | Requirement reference | Test 1/ method | Sample unit |
|---|---|---|---|---|
| Pile yarn (lbs.) - | Material identification. | 3.4.1 | 2/ | |
| | Ply. | 3.4.1 | Visual | |
| | Turns per inch (type I). | 3.5.1 | 4054 | ½ lb. |
| | Chloroform soluble material. | 3.4.1.1 | 2611 | |
| | Wool content. | 3.4.1.2 | 2101 | |
| Backing materials (yds.) | | | | |
| 1st and 2nd backings (type I) | Material identification. | 3.4.2.1 | 2/ | ½ yd. (for each backing material) |
| | Weight. | 3.4.2.1 | 5041 | |
| base cloth (type II) | Material identification. | 3.4.2.2 | 2/ | |
| | Weight. | 3.4.2.2 | 5041 | ½ yd. |
| | Yarns per inch. | 3.4.2.2 | 5050 | |
| solid backing (type II) | Material identification. | 3.4.2.1 and 3.4.2.2 | 2/ | |
| | Thickness. | 3.4.2.1 and 3.4.2.2 | 2121 3/ | ½ yd. |
| | Accelerated weathering (class 1). | 3.4.2.1 | 5804 | |
| | Accelerated aging (class 2). | 3.4.2.2 | 7221 or 7241 4/ | |
| Back coating compound (pt.) | Material identification. | 3.4.3 | 2/ | ½ pt. |
| Binding tape (type I)(yds.) | Material identification. | 3.5.1.1 | 2/ | |
| | Width. | 3.5.1.1 | Visual | 1 yd. |
| | Number of ends. | 3.5.1.1 | 5050 | |
| | Number of picks. | 3.5.1.1 | 5050 | |
| | Weight. | 3.5.1.1 | 5041 5/ | |

1/ Except as otherwise noted, test methods referenced are specified in Federal Specification CCC-T-191.

2/ May be accepted based on a supplier's certificate of compliance.

3/ of Fed. Test Method Std. No. 501.

4/ of Fed. Test Method Std. No. 601.

5/ Except that the weight shall be reported to the nearest 0.01 ounce.



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TABLE IV - Sample size

| Lot size (units) | Sample size |
|---|---|
| 800 or less | 2 |
| 801 to 22,000 | 3 |
| 22,001 and over | 5 |

4.3.2 Examination of the end item. The mats or matting shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.3.2.1, 4.3.2.2 and 4.3.2.3.

4.3.2.1 Visual examination. The defects listed in table V shall be counted regardless of their proximity to one another, except where two or more defects represent a single local condition of the mat or matting, in which case only the more serious defect shall be counted. When examining rolls of matting, continuous defects shall be counted as one defect for each yard or fraction thereof in which they occur. The sample unit shall be 1 linear yard when matting in rolls are furnished and one mat when individual mats are furnished. The acceptable quality levels shall be 1.5 major and 6.5 total defects (major and minor combined) per hundred units. The inspection level shall be level II. The lot size shall be expressed in units of 1 linear yard when matting in rolls are furnished and one mat when individual mats are furnished. When examination is made of matting furnished in rolls, not more than one third of the total sample yardage shall be examined in any one roll. When the lot consists of less than 5 rolls, an approximate equal number of yards shall be examined in each roll to yield the sample yardage.

TABLE V - Classification of defects

| Examine | Defects | Classification Major | Minor |
|---|---|---|---|
| Material and workmanship | Any hole or tear. | X | |
| | Tufts missing in pile. | | X |
| | Any specified component missing. | X | |
| | Not constructed as specified. | X | |
| | Back coating skips (type I). | | X |
| | Edges not bound as specified (type I). | X | |
| | Backing reinforcement incorrectly applied, with wrinkles or poor adherence (type I.) | | X |
| | Pile yarn or base cloth not firmly secured by fusing, curing or vulcanizing, as applicable, to backing (type II). | X | |
| | Tufts easily pulled from pile surface. | X | |
| | Obvious curling of nosing or borders. | X | |
| | Nosing insecurely attached or obviously of improper height which could cause tripping by pedestrians. | X | |
| Finish and appearance | Spot or stain. | | X |
| | Any clearly obvious streak lengthwise of weave. | X | |
| | Discolored areas affecting appearance (variations in color due to crushing of pile shall not be considered a defect). | | X |
| | Any evidence of migration of color from or color rubbing off backings. | X | |
| | Marks across mat or matting. | | X |
| | Length of mats or width not as specified. | | X |
| | Incorrect color. | | X |
| | Nosing or borders distorted. | X | |
| | Any noticeable unevenness on top surface affecting appearance. | | X |



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4.3.2.2 Overall examination. Each defect listed shall be counted not more than once in each unit examined. The sample unit for this examination shall be one roll for matting or one mat when individual mats are furnished. The number of rolls or mats examined for visual examination shall be the sample size for this examination. The lot shall be unacceptable if one or more of the following defects are found.

Defects

Overall uncleanness.
Color other than specified.
Rancid or otherwise objectionable odor.
Type not as specified.
Edges not finished as required.
Not constructed as required.

4.3.2.3 Length examination (matting). Each roll of matting shall be examined for gross length. Any gross length found to be more than 1 inch less than the gross length marked on the roll ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The rolls used for the visual examination of matting shall be used for the examination. The lot shall be unacceptable if one or more defects are found in the sample.

4.3.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VI. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE VI - Classification of preparation for delivery defects.

| Examine | Defects |
|---|---|
| Markings (exterior and interior) | Omitted; incorrect; illegible; improper size, location, sequence, or method of application. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of wrap. Securing of wrap inadequate. |

4.4 Testing of the end item. Tests shall be performed for the characteristics specified in tables VII and VIII, as applicable. Unless otherwise indicated, the physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be the quantity of material required to perform all the tests listed in tables VII and VIII, as applicable. The sample size shall be as specified in table IV. The lot size for purposes of end item testing shall be expressed in units of linear yards for rolls of matting and mats when individual mats are furnished. Except for the requirements for weight of the back coating for type I mats or matting, the lot shall be unacceptable if one or more sample units fail to meet any specified requirement. For the weight of the back coating (type I), the lot shall be unacceptable if the lot average fails to meet the specified requirement.



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TABLE VII - End item testing (type I)

| Characteristic | Requirement references | Test method |
|---|---|---|
| Tufts, square inch | Table I | 4.5.1 |
| Weight, pile yarn | Table I | 4.5.2.1 |
| Weight, overall | Table I | 5040 or 5041 1/ 2/ |
| Pile height | Table I | 4.5.4.1 |
| Weight of back coating 3/ | 3.5.1 | 4.5.5 |
| Resistance to wet method cleaning | Table I | 4.5.7 |
| Resistance to curl | 3.5.2 and 3.6.2 | 4.5.12 |
| Flame resistance | Table I | 4.5.8 |
| Colorfastness to light | 3.8.1 | 5660 1/ 4/ |
| Colorfastness to wet-method cleaning | 3.8.1 | 4.5.10 |
| Colorfastness to water | 3.8.1 | 4.5.11 |
| Shrinkage | 3.9 | 4.5.9 |
| Moth repellency (as applicable) | 3.10 | ASTM D-1116-55T |

1/ of Federal Specification CCC-T-191
2/ When method 5041 is used, one specimen with an area of at least 16 square inches shall be used.
3/ The test results for this characteristic shall be reported as the lot average.
4/ In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer shall authorize the exposure to natural light in accordance with method 5662 of CCC-T-191.

TABLE VIII - End item testing (type II, classes 1 and 2).

| Characteristic | Requirement references | Test method |
|---|---|---|
| Tufts, square inch | Table II | 4.5.1 |
| Weight, pile yarn 1/ | Table II | 4.5.2.2 2/ |
| Weight, overall | Table II | 5040 or 5041 3/ 4/ |
| Thickness, overall | Table II | 4.5.3 |
| Pile height | Table II | 4.5.4.2 |
| Flexibility: | | |
| 70° F. | Table II | 4.5.6 |
| 30° F. | Table II | 6511 5/ |
| Resistance to wet method cleaning | Table II | 4.5.7 |
| Flame resistance | Table II | 4.5.8 6/ |
| Colorfastness to light | 3.8.1 | 5660 3/ 7/ |
| Colorfastness to wet-method cleaning | 3.8.1 | 4.5.10 |
| Colorfastness to water | 3.8.1 | 4.5.11 |
| Contact and migration stain (when applicable) | 3.8.2 | ASTM D-925-63 8/ |
| Moth repellency (as applicable) | 3.10 | ASTM D-1116-55T |

1/ May be accepted based on contractor's Certificate of Compliance.
2/ In the event of dispute resulting from the test method specified, the pile yarn shall be separated from the backing on a 1 inch by 1 inch sample and the pile weight determined accordingly. The same number of sample units shall be tested as that established for the test listed above.
3/ of Federal Specification CCC-T-191.
4/ When method 5041 is used, one specimen with an area of at least 16 square inches shall be used.
5/ of Fed. Test Method Std. No. 501. The diameter of the mandrel shall be $\frac{1}{4}$(± 1/16) inch.
6/ The charred area of the backing shall be included in the measurement.
7/ In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer shall authorize the exposure to natural light in accordance with method 5662 of CCC-T-191.
8/ The contact and migration stain test shall be conducted only when specified in the contract or order. The organic finishes or flooring shall be specified in the contract or order.

4.5 Tests.

4.5.1 Tufts, square inch. Tufts per square inch shall be determined by multiplying the pitch per inch of pile yarn ends by the rows of pile per inhh, or the equivalent thereof. Pitch and rows per inch shall be determined in accordance with ASTM D-418-42. Pitch is defined as the number of pile yarn ends per inch of width in the fabric. Rows is defined as the number of rows of tufts per inch of length (also referred to as stitches per inch for tufted).



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4.5.2 Weight of pile yarn.

4.5.2.1 Type I. The pile yarn shall be separated from other yarns on a sample containing 16 square inches. To express weight of pile yarn for mat or matting (36" by 36"), the weight of the pile shall be calculated as follows: Weight of pile yarn (ounces) in 16 square inch sample X81.0= weight of pile yarn in ounces per square yard of mat or matting.

4.5.2.2 Type II, classes 1 and 2. The pile yarn weight shall be determined by subtracting the known backing and base cloth weight from the total weight of the mat or matting on a 6 by 6 inch sample. The difference shall be the pile yarn weight.

4.5.3 Thickness, overall (type II, classes 1 and 2). The thickness shall be determined by placing the sample on a flat and level surface and measuring the floor covering height with a metal rule graduated in 1/100 inches or 1/64 inches approximately 3/4 inch wide and .040 inch thick. Five measurements are to be made at 2.0 inch intervals by placing the rule on the surface beside the sample and taking readings. The average of the five readings shall be considered the mat or matting thickness.

4.5.4 Pile height.

4.5.4.1 Type I. The pile height shall be determined in accordance with ASTM D-1486-57T for type I mats or matting.

4.5.4.2 Type II, classes 1 and 2. The same procedure shall be followed as specified in 4.5.3, except that the rule shall be placed on the top of base cloth along edge. Brush up pile before taking measurements. The average of the five readings shall be considered the pile height.

4.5.5 Weight of back coating (type I). The weight of the mat or matting after back coating shall be determined in accordance with method 5040 or 5041 of CCC-T-191. When method 5041 is used, one specimen with an area of at least 16 square inches shall be used. The results of this test shall be reported as the lot average. The individual sample unit results obtained for the total weight characteristics (exclusive of back coating) shall be averaged and this result subtracted from the lot average weight after back coating. The difference obtained shall be the lot average weight of the back coating and shall be reported to the nearest ounce.

4.5.6 Flexibility at 70° F. (type II, classes 1 and 2). A sample of the mat or matting shall be bent 170 degrees over a ½(± 1/16) inch mandrel, first bending the backside and then bending the face or pile side, and continuing this alternate bending for a total of 20 bends. The test shall be conducted in an area having a temperature of 70°(±2°)F.

4.5.7 Resistance to wet-method cleaning. A 12 by 12 inch sample of the mat or matting shall be conditioned under standard conditions as defined in CCC-T-191 for a period of 24 hours. The specimen shall be marked and measured at three different locations in the length and width directions; immerse sample in 110° F. water for 15 minutes; mix 2 g. of sodium alkylsulfate type of detergent with 50 g. of water at 110° F. and apply to pile surface of mat or matting. Scrub sample with a clean soft bristle brush by stroking back and forth 20 times (10 times in each unilateral direction), and in both length and width directions. Rinse well to remove majority of detergent and dry at air temperature. Scrub sample again, same as above, using a clean soft bristle brush. A 12 by 12 inch control sample (not subjected to detergent or water) shall be scrubbed, same as above, using a clean soft bristle brush. The amount of pile yarn or fiber picked up either by the brush on the sample with the detergent or by the brush on the sample after rinsing shall be not more than that picked up by the control sample. The pressure used during the scrubbing shall be the same for the three samples and shall be approximately equal to that the mats or matting would be subjected to in usage. Note that type I samples used in this test are remeasured for shrinkage as specified in 4.5.9.

4.5.8 Flame resistance. This method covers the procedure for measuring flammability. The test shall be carried out in a suitably sized chamber or box which is open at the top to provide adequate ventilation and which, at the same time, protects the specimen and the ignition flame from drafts. The box should be 12 inches by 12 inches by 9 inches. The ignition media shall be Methenamine Timed Burning Tablet (Eli Lilly #1588). The measuring device shall be a scale graduated in inches and tenths of inches. Two test specimens, each 6 inches square, shall be cut from the end item to be tested. The test shall be done using 30 percent relative humidity. This condition is to be arrived at by using a small laboratory drying oven in conjunction with the standard conditioned testing room. From the psychrometric chart, it can be determined that air at 70° F. and 65 percent relative humidity will have a relative humidity of 30 percent when heated to 94° F. Set the oven controls to maintain this latter temperature, place the samples in the oven and allow them to come to equilibrium. Without disturbing the configuration of the pile, the sample shall then be laid out in the test chamber smoothly, horizontally and without



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tension. A burning tablet shall then be placed firmly in the center of the sample and ignited by touching a match carefully to the edge of the tablet so as not to contact the surface of the mat or matting. The ignition flame and any propagated flame shall be allowed to burn to completion. The longest diameter of the charred area shall be measured to the nearest tenth of an inch.

4.5.9 Shrinkage (type I). After completion of the test specified in 4.5.7, the cleaned type I samples used for the test shall be remeasured and the shrinkage computed using the following formula:

$$\text{Shrinkage} = \frac{A - B}{A} \times 100$$

where:
A = Average of initial measurement.
B = Average measurements after shampooing.

Note: The shrinkage in both the length and width directions shall be reported to the nearest 0.1 percent.

4.5.10 Colorfastness to wet-method cleaning. After completion of the test specified in 4.5.7, the cleaned samples shall be used for colorfastness evaluation. Evaluation shall be conducted in accordance with method 5610 of CCC-T-191.

4.5.11 Colorfastness to water. A 6 by 6 inch sample of the mat or matting shall be immersed in 140° F. water for 60 minutes. Rinse well and dry at 125° F. until dry. Colorfastness evaluation shall be conducted in accordance with method 5610 of CCC-T-191.

4.5.12 Resistance to curl. Samples of the finished mat shall be laid on the floor. There shall be no curling condition, in that the mats shall lay flat on the floor at all points around the edges, including nosing and borders. In addition, the edges of the finished mat shall be bent or folded by hand ten times (to simulate kicking or pushing up of the edge as might occur in normal traffic). The mat should revert to natural position, i. e. flat on the floor at all points.

4.5.13 Color matching test. A test specimen at least 8 inches square shall be mounted at an angle 45° to the horizontal and compared with the standard sample, or the approved color sample furnished by the contracting officer, similarly mounted under the light conditions specified in 3.8. When an artificial light source is used it shall be placed so that its rays strike the sample normal to its surface. The viewing distance shall be 18 to 24 inches.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Matting. Rolls of matting furnished in accordance with 3.7 in widths and lengths as specified (see 6.2) shall be wrapped with two thicknesses of material conforming to PPP-B-1055, class E-2. The seams shall be secured with adhesive conforming to MMM-A-260 with the ends and folds secured with 2-1/2 inch wide tape conforming to PPP-T-76 effecting a watertight seal.

5.1.1.2 Mats. The mats furnished in special cut sizes in accordance with 3.7 in widths and lengths as specified (see 6.2)shall be wrapped with two thicknesses of material conforming to PPP-B-1055, class E-2. The seams shall be secured with adhesive conforming to MMM-A-260 with the ends and folds secured with 2-1/2 inch wide tape conforming to PPP-T-76 effecting a watertight seal.

5.1.2 Level B.

5.1.2.1 Matting. Rolls of matting furnished in accordance with 3.7 in widths and lengths as specified (see 6.2) shall be wrapped with kraft paper reinforced with burlap. The seams, ends and folds, shall be secured with 2-1/2 inch wide tape conforming to PPP-T-45, type I, class 2.

5.1.2.2 Mats. The mats furnished in special cut sizes as specified in 3.7 in widths and lengths as specified (see 6.2) shall be wrapped with kraft paper reinforced with burlap. The seams, ends and folds, shall be secured with 2-1/2 inch wide tape conforming to PPP-T-45.



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5.1.3 Level C. The matting or mats shall be packaged to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The rolls of matting or special cut sizes of mats of the same type, class or color shall be packed in a close-fitting box conforming to PPP-B-601, overseas type or PPP-B-621, class 2. The box shall be closed and strapped in accordance with the appendix to the box specification. When the gross weight exceeds 250 pounds, skids shall be provided as specified in the applicable specification.

5.2.2 Level B. No packing required.

5.2.3 Level C. The matting or mats shall be packed to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The mats and matting covered by this specification are intended for use in entrance ways, corridors, lobbies, elevators and similar areas to reduce tracking of dirt, water, snow and grit into the interior of buildings, to help prevent slips and falls, to protect floors and to reduce noise.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class, as applicable, required (1.2.1).
(c) When preproduction sample is required (3.1).
(d) When bid samples are required (3.3).
(e) Pile fiber required (3.4.1).
(f) Binding of edges for type I, if other than as specified (3.5.1.1).
(g) Pile surface (cut or loop), if other than as specified (3.5.2).
(h) Specify nosing required for type I (3.6.1).
(i) For type II, specify borders required if other than as specified or nosing when required (3.6.2).
(j) Size (width and length) required (3.7).
(k) Colors required (3.8 and 6.4).
(l) When contact and migration stain test is required; specify requirement and organic finish or flooring applicable (3.8.2 and table VIII).
(m) Moth repellency requirement, if other than as specified (3.10).
(n) Selection of applicable level of packaging and packing required (5.1 and 5.2).

6.3 Guidelines for selection of pile surface mats or matting. As a guide in determining the feasibility of using pile surface mats for special purposes, such as to reduce tracking of dirt, grit, water and snow into the interior of buildings, the following guidelines are listed. It should be kept in mind that each installation must be judged carefully as to the peculiar traffic, environmental and climatic conditions expected at the installation site. It is emphasized that these guidelines apply to special purpose pile surface mats, such as the mats covered by this specification, and do not necessarily apply to carpeting of conventional construction (e.g. Velvet, Axminster, etc.) for interior use (e.g. in offices, etc.) where traffic and appearance only are the primary considerations. The guidelines are:

a. Pile surface floor covering for entranceway type use should be of the "pick-up" type for easy cleaning, drying, and general maintenance and rotation when exposed to heavy traffic and/or climatic or environmental conditions resulting in substantial accumulation and tracking of water and snow and other contaminants such as dirt, grit and chemicals. Under these conditions the pile surface floor covering is not intended for permanent or fixed type



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of installation, where it cannot be readily removed to dry out or otherwise be properly maintained.

b. Cigarette disposal at the installation site should be controlled. Lighted cigarette butts thrown on the pile surface can cause some damage to the floor covering.

c. Pile height should be considered so that doors will swing freely over the floor coverigg.

d. As an overall guide for users in the selection of pile surface mats for use at entranceways and similar areas to resist tracking, consideration should be given to the performance expected from the mats at the particular installation site as related to the following characteristics:

1. Resistance to tracking of dirt and grit.
2. Resistance to tracking of water and snow.
3. Resistance to tracking of chemicals (e.g. calcium chloride).
4. Resistance to formation of puddles on mat surface.
5. Whether non-skidding (mat on floor).
6. Whether non-slipping (to prevent slipping of pedestrians on surface of mat).
7. Correct pile height (e.g. to allow for closing of doors).
8. Appearance level (including color).
9. Non-curling (to prevent tripping by pedestrians-e.g. has nosing, etc.).
10. Availability of proper cleaning and maintenance facilities.

6.4 Colors. For the guidance of purchasers, dark colors are most satisfactory for pile surface mats used at entranceway type areas, including black, brown, and wine in moresque, tweed or solid colors.

6.5 Standard sample. For access to the standard sample, when available, address the procuring office issuing the invitation for bids.



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